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Table: 10.5 Types of receptors

Receptors	Stimulus	Effector organs
Mechano receptors	Pressure and vibration	Mechano receptors are present in the cochlea of the inner ear and the semi circular canal and utriculus
Chemoreceptors	Chemicals	Taste buds in the tongue and nasal epithelium
Thermoreceptors	Temperature	Skin
Photoreceptors	Light	Rod and cone cells of the retina in the eye

10. 6 Sensory reception and processing

Our senses make us aware of changes that occur in our surroundings and also within our body. **Sensation** [awareness of the stimulus] and **perception** [interpretation of the meaning of the stimulus] occur in the brain.

Receptors are classified based on their location: **1. Exteroceptors** are located at or near the surface of the body. These are sensitive to external stimuli and receive sensory inputs for hearing, vision, touch, taste and smell. **2. Interoceptors** are located in the visceral organs and blood vessels. They are sensitive to internal stimuli. **Proprioceptors** are also a kind of interoceptors. They provide information about position and movements of the body. These are located in the skeletal muscles, tendons, joints, ligaments and in connective tissue coverings of bones and muscles. Receptors based on the type of stimulus are shown in Table 10.5.

10.6.1 Photoreceptor - Eye

Eye is the organ of vision; located in the orbit of the skull and held in its position with the help of six extrinsic muscles. They are **superior, inferior, lateral, median rectus muscles, superior oblique and inferior oblique** muscles. These muscles aid in the movement of the eyes and they receive their nerve innervation from III, IV and VI cranial nerves. Eyelids, eye lashes and eye brows are the accessory structures useful in protecting the eyes. The eye lids protect the eyes from excessive light and foreign objects and spread lubricating secretions over the eyeballs.

Eyelashes and the eyebrows help to protect the eyeballs from foreign objects, perspiration and also from the direct rays of sunlight. **Sebaceous glands** at the base of the eyelashes are called **ciliary glands** which secrete a lubricating fluid into the hair follicles. **Lacrymal glands,** located in the upper lateral region of each orbit, secrete tears. Tears are secreted at the rate of 1mL/day and it contains salts, mucus and **lysozyme** enzyme to destroy bacteria.

The conjunctiva is a thin, protective mucous membrane found lining the outer surface of the eyeball (Figure 10.14).

The eye has two compartments, the **anterior** and **posterior compartments**. The anterior compartment has two chambers,

Your friend is returning home after his visit to USA. All at home are waiting for his arrival. How would you feel? State the division of ANS that predominates and mention few changes that take place in your body.